**EXP 7 : Analysis of 5G NR Frame Structure: Resource Grid, Subcarrier Spacing and RAN Functions.**

**MATLAB CODE:**

```
clc;
clear;
close all;

%% 5G NR FRAME STRUCTURE ANALYSIS

% 5G NR numerology
SCS = [15 30 60 120];       % Subcarrier spacing in kHz
mu = [0 1 2 3];

% Frame parameters
frameDuration = 10;         % Frame duration in ms
subframeDuration = 1;       % Subframe duration in ms
symbolsPerSlot = 14;        % Normal cyclic prefix
RB = 24;                    % Number of resource blocks

%% 1. NUMEROLOGY CALCULATION

slotsPerSubframe = 2.^mu;
slotDuration = 1./(2.^mu);

totalSlots = slotsPerSubframe*10;

fprintf('\n--- 5G NR NUMEROLOGY ---\n');

for i = 1:length(SCS)

   fprintf('\nSubcarrier Spacing: %d kHz\n',SCS(i));

   fprintf('Slots per Subframe: %d\n', ...
      slotsPerSubframe(i));

   fprintf('Slot Duration: %.3f ms\n', ...
      slotDuration(i));

   fprintf('Slots per Frame: %d\n', ...
      totalSlots(i));
```

```
    fprintf('Symbols per Slot: %d\n', ...
        symbolsPerSlot);

end

%% 2. 5G NR RESOURCE GRID

% 12 subcarriers per resource block
totalSubcarriers = RB*12;

% Resource grid:
% 0 = Empty resource
% 1 = Control channel
% 2 = Data channel
% 3 = Reference signal

resourceGrid = zeros(totalSubcarriers,symbolsPerSlot);

% Allocate control resources
resourceGrid(1:24,1:2) = 1;

% Allocate data resources
resourceGrid(25:200,3:12) = 2;

% Allocate reference signals
resourceGrid(10:10:end,4:14) = 3;

figure('Name','5G NR Resource Grid');

imagesc(resourceGrid);

colormap([1 1 1;
          0.2 0.4 0.9;
          0.2 0.8 0.3;
          1 0.5 0.1]);

colorbar;

xlabel('OFDM Symbols');
ylabel('Subcarriers');

title('5G NR Time-Frequency Resource Grid');
```

```
xticks(1:14);
grid on;

%% 3. SUBCARRIER SPACING COMPARISON

figure('Name','Subcarrier Spacing');

bar(SCS);

set(gca,'XTickLabel', ...
    {'15 kHz','30 kHz','60 kHz','120 kHz'});

xlabel('5G NR Numerology');
ylabel('Subcarrier Spacing (kHz)');

title('Comparison of Subcarrier Spacing');

grid on;

%% 4. SLOT DURATION COMPARISON

figure('Name','Slot Duration');

bar(SCS,slotDuration);

xlabel('Subcarrier Spacing (kHz)');
ylabel('Slot Duration (ms)');

title('5G NR Slot Duration Comparison');

grid on;

%% 5. RESOURCE GRID FOR DIFFERENT NUMEROLOGIES

figure('Name','Resource Grid Comparison');

for i = 1:4

    subplot(2,2,i);

    % Create resource grid
    gridData = zeros(12*RB,14);
```

```
    % Control allocation
    gridData(1:24,1:2) = 1;

    % Data allocation
    gridData(25:end,3:14) = 2;

    % Reference signal allocation
    gridData(5:12:end,4:14) = 3;

    imagesc(gridData);

    title(['SCS = ' num2str(SCS(i)) ' kHz']);

    xlabel('OFDM Symbols');
    ylabel('Subcarriers');

end

sgtitle('5G NR Resource Grid for Different Numerologies');

%% 6. FRAME STRUCTURE

figure('Name','5G NR Frame Structure');

hold on;

% Draw 10 subframes
for i = 0:9

    rectangle('Position',[i,0,1,1], ...
        'FaceColor',[0.2 0.6 0.9], ...
        'EdgeColor','black');

    text(i+0.5,0.5,['SF ' num2str(i)], ...
        'HorizontalAlignment','center', ...
        'Color','white','FontWeight','bold');

end

xlim([0 10]);
ylim([0 1.5]);
```

```
xlabel('Time (ms)');
ylabel('Frame');

title('5G NR Frame Structure (10 ms Frame)');

text(5,1.2,'10 Subframes x 1 ms = 10 ms Frame', ...
    'HorizontalAlignment','center');

grid on;
hold off;

%% 7. RAN FUNCTIONS

figure('Name','5G RAN Architecture');

axis off;
hold on;

% User Equipment
rectangle('Position',[0.5 2 2 1], ...
    'FaceColor',[0.8 0.9 1]);

text(1.5,2.5,'User Equipment (UE)', ...
    'HorizontalAlignment','center');

% Radio Unit
rectangle('Position',[3.5 2 2 1], ...
    'FaceColor',[0.8 1 0.8]);

text(4.5,2.5,'Radio Unit (RU)', ...
    'HorizontalAlignment','center');

% Distributed Unit
rectangle('Position',[6.5 2 2 1], ...
    'FaceColor',[1 0.9 0.7]);

text(7.5,2.5,'Distributed Unit (DU)', ...
    'HorizontalAlignment','center');

% Centralized Unit
rectangle('Position',[9.5 2 2 1], ...
    'FaceColor',[1 0.8 0.8]);
```

```
text(10.5,2.5,'Central Unit (CU)', ...
    'HorizontalAlignment','center');

% Connections
plot([2.5 3.5],[2.5 2.5],'k','LineWidth',2);
plot([5.5 6.5],[2.5 2.5],'k','LineWidth',2);
plot([8.5 9.5],[2.5 2.5],'k','LineWidth',2);

text(3,2.8,'Radio');
text(6,2.8,'Fronthaul');
text(9,2.8,'Midhaul');

title('5G Radio Access Network Architecture');

xlim([0 12]);
ylim([1 4]);

hold off;

%% 8. RAN FUNCTION DESCRIPTION

fprintf('\n--- 5G RAN FUNCTIONS ---\n');

fprintf('\n1. USER EQUIPMENT (UE)\n');
fprintf('Mobile device communicates with the 5G network.\n');

fprintf('\n2. RADIO UNIT (RU)\n');
fprintf('Handles RF transmission and reception.\n');

fprintf('\n3. DISTRIBUTED UNIT (DU)\n');
fprintf('Handles real-time baseband processing.\n');

fprintf('\n4. CENTRAL UNIT (CU)\n');
fprintf('Handles higher-layer protocol processing.\n');

%% 9. FINAL COMPARISON TABLE

Results = table( ...
    SCS', ...
    slotsPerSubframe', ...
    slotDuration', ...
    totalSlots', ...
    'VariableNames', ...
```

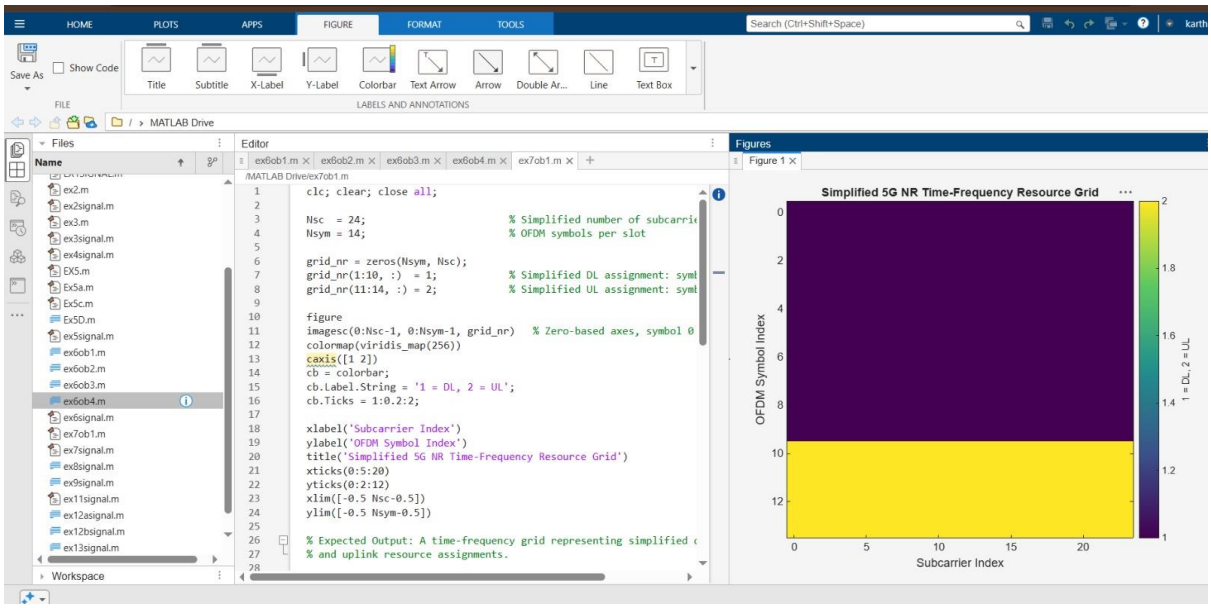
```
    {'SCS_kHz','Slots_Per_Subframe', ...
    'Slot_Duration_ms','Slots_Per_Frame'});

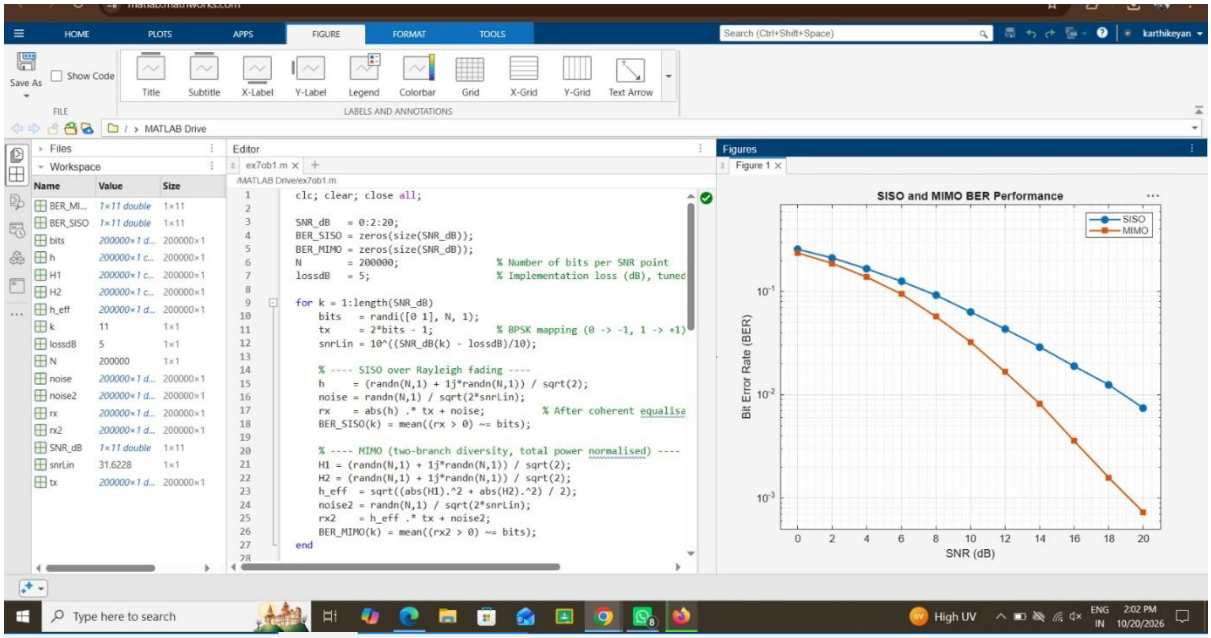
disp(' ');
disp('--- FINAL 5G NR COMPARISON TABLE ---');

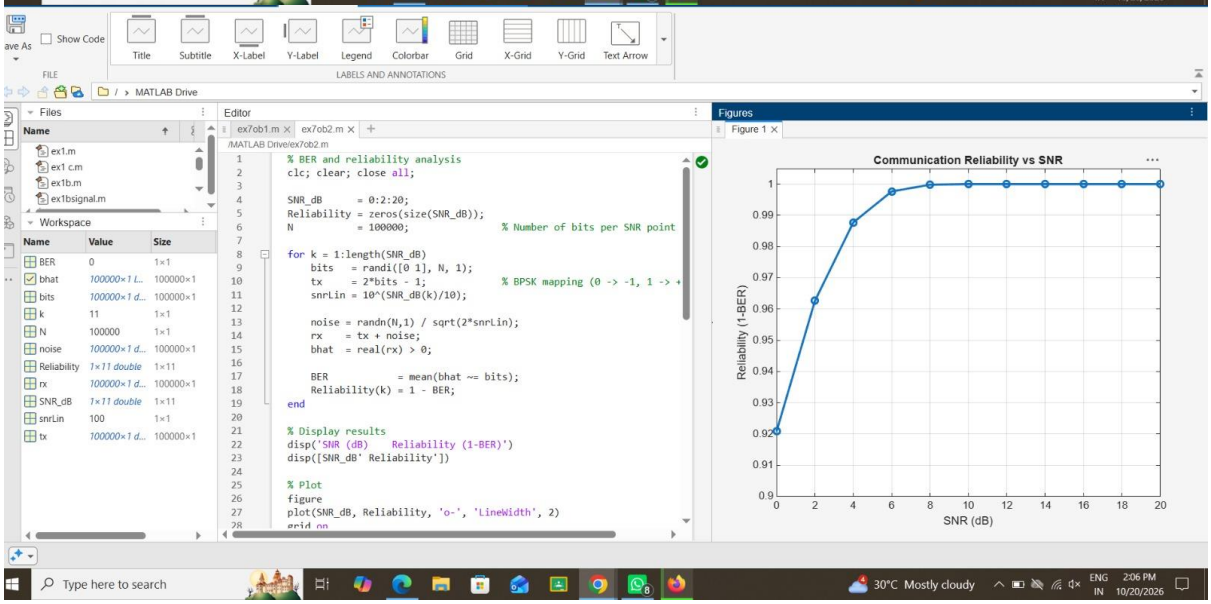
disp(Results);

fprintf('\n5G NR Simulation Completed Successfully!\n');
```

OUTPUT:




HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
karthikeyan
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
Text Arrow
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Workspace
Name
Value
Size
BER_MI... 1×11 double 1×11
BER_SISO 1×11 double 1×11
bits 200000×1 d... 200000×1
h 200000×1 c... 200000×1
H1 200000×1 c... 200000×1
H2 200000×1 c... 200000×1
h_eff 200000×1 d... 200000×1
k 11 1×1
lossdB 5 1×1
N 200000 1×1
noise 200000×1 d... 200000×1
noise2 200000×1 d... 200000×1
rx 200000×1 d... 200000×1
rx2 200000×1 d... 200000×1
SNR_dB 1×11 double 1×11
snrLin 31.6228 1×1
tx 200000×1 d... 200000×1
Editor
ex7ob1.m
/MATLAB Drive/ex7ob1.m
clc; clear; close all;
SNR_dB = 0:2:20;
BER_SISO = zeros(size(SNR_dB));
BER_MIMO = zeros(size(SNR_dB));
N = 200000; % Number of bits per SNR point
lossdB = 5; % Implementation loss (dB), tuned
for k = 1:length(SNR_dB)
bits = randi([0 1], N, 1);
tx = 2*bits - 1; % BPSK mapping (0 -> -1, 1 -> +1)
snrLin = 10^((SNR_dB(k) - lossdB)/10);
% ---- SISO over Rayleigh fading ----
h = (randn(N,1) + 1j*randn(N,1)) / sqrt(2);
noise = randn(N,1) / sqrt(2*snrLin);
rx = abs(h) .* tx + noise; % After coherent equalisa
BER_SISO(k) = mean((rx > 0) ~= bits);
% ---- MIMO (two-branch diversity, total power normalised) ----
H1 = (randn(N,1) + 1j*randn(N,1)) / sqrt(2);
H2 = (randn(N,1) + 1j*randn(N,1)) / sqrt(2);
h_eff = sqrt((abs(H1).^2 + abs(H2).^2) / 2);
noise2 = randn(N,1) / sqrt(2*snrLin);
rx2 = h_eff .* tx + noise2;
BER_MIMO(k) = mean((rx2 > 0) ~= bits);
end
Figures
Figure 1
SISO and MIMO BER Performance
SISO
MIMO
Bit Error Rate (BER)
SNR (dB)
Type here to search
High UV
ENG
IN
2:02 PM
10/20/2026


Show Code
Title
Subtitle
X-Label
Y-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
Text Arrow
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Files
ex1.m
ex1 c.m
ex1b.m
ex1bsignal.m
Workspace
Name
Value
Size
BER 0 1×1
bhat 100000×1 l... 100000×1
bits 100000×1 d... 100000×1
k 11 1×1
N 100000 1×1
noise 100000×1 d... 100000×1
Reliability 1×11 double 1×11
rx 100000×1 d... 100000×1
SNR_dB 1×11 double 1×11
snrLin 100 1×1
tx 100000×1 d... 100000×1
Editor
ex7ob1.m
ex7ob2.m
/MATLAB Drive/ex7ob2.m
% BER and reliability analysis
clc; clear; close all;
SNR_dB = 0:2:20;
Reliability = zeros(size(SNR_dB));
N = 100000; % Number of bits per SNR point
for k = 1:length(SNR_dB)
bits = randi([0 1], N, 1);
tx = 2*bits - 1; % BPSK mapping (0 -> -1, 1 -> +
snrLin = 10^(SNR_dB(k)/10);
noise = randn(N,1) / sqrt(2*snrLin);
rx = tx + noise;
bhat = real(rx) > 0;
BER = mean(bhat ~= bits);
Reliability(k) = 1 - BER;
end
% Display results
disp('SNR (dB) Reliability (1-BER)')
disp([SNR_dB' Reliability'])
% Plot
figure
plot(SNR_dB, Reliability, 'o-', 'LineWidth', 2)
grid on
Figures
Figure 1
Communication Reliability vs SNR
Reliability (1-BER)
SNR (dB)
Type here to search
30°C Mostly cloudy
ENG
IN
2:06 PM
10/20/2026